TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	82.6 ± 0.92
Correlation Stability	82.2 ± 0.60
Distribution Consistency	85.4 ± 1.02
Entropy Preservation	86.2 ± 1.08
Mean Z Score Drift	0.3 ± 0.08
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	84.5 ± 3.72
Novelty Score	0.9 ± 0.01

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	53.409	-1.030	1.9%	OK
sex	0.680	0.682	+0.002	0.3%	OK
cp	3.158	3.269	+0.110	3.5%	OK
trestbps	131.690	123.331	-8.359	6.3%	OK
chol	246.693	244.479	-2.214	0.9%	OK
fbs	0.149	0.236	+0.087	58.6%	Drift
restecg	0.990	1.099	+0.109	11.0%	Warn
thalach	149.607	156.661	+7.054	4.7%	OK
exang	0.327	0.339	+0.012	3.7%	OK
oldpeak	1.040	1.222	+0.182	17.5%	Warn
slope	1.601	1.694	+0.094	5.8%	OK
ca	0.672	0.818	+0.146	21.7%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	82.6 ± 0.92
Correlation Stability (*2)	82.2 ± 0.60
Distribution Consistency (*3)	85.4 ± 1.02
Entropy Preservation (*4)	86.2 ± 1.08
Mean Z Score Drift (*5)	0.3 ± 0.08
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	84.5 ± 3.72
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5344262295081966	0.5770491803278689	-0.02131147540983607	
F1	0.5648347697855012	0.3967764434076343	0.48175268578582486	-0.08308208399967629	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5262295081967212	0.3934426229508197	0.5081967213114755	-0.01803278688524591	
F1	0.5272786716544238	0.3709339385465748 6	0.4970818112440892	-0.03019686041033447	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.4491803278688525	0.5672131147540984	0.02459016393442624	
F1	0.5198013329305267	0.3865406507361028	0.5332327035243877	0.013431370593860842	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5360655737704919	0.4573770491803278 4	0.5688524590163935	0.03278688524590164	
F1	0.5231512298313296	0.3974341698737429	0.5354436568693873	0.012292427038057602	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.5180327868852459	0.5819672131147541	-0.02622950819672131	
F1	0.6070236204175147	0.4248705718734934	0.5232394017111373	-0.0837842187063774	*

Model: KNN

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.5147540983606558	0.5245901639344261	-0.02950819672131147	
F1	0.4442202071436568 4	0.3888636552478867 4	0.4152512317043269	-0.02896897543932994	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.4508196721311474 3	0.5934426229508196	0.03606557377049182	
F1	0.568660979990255	0.3831080336096369	0.5627073544696674	-0.00595362552058761 35	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5180327868852459	0.4590163934426229	0.5672131147540982	0.04918032786885246 5	
F1	0.4782496199381977	0.3927619156254107 4	0.4805753210878501 7	0.00232570114965253 72	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5475409836065573	0.5475409836065573	0	
F1	0.3899270600110827	0.3899270600110827	0.3899270600110827	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5901639344262295	0.5245901639344261	0.5967213114754099	0.00655737704918035 7	
F1	0.5480779520963047	0.3929649535682405 7	0.5223182328071901	-0.02575971928911454	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	47.0	1.0	4.0	161.0	216.0	0.0	0.0	100.0	0.0	-0.006597058108270382
2	47.0	1.0	4.0	111.0	313.0	0.0	0.0	122.0	0.0	0.7868157968063512
3	62.0	1.0	3.0	134.0	299.0	1.0	2.0	126.0	0.0	0.05379072591568155
4	48.0	1.0	2.0	98.0	184.0	0.0	0.0	123.0	0.0	-0.03388984029896491
5	63.0	1.0	2.0	138.0	254.0	0.0	1.0	192.0	0.0	-0.14872612530543394
6	54.0	1.0	2.0	125.0	206.0	0.0	2.0	187.0	0.0	0.27341464886437944
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	42.0	1.0	3.0	123.0	131.0	0.0	2.0	163.0	1.0	3.5642923419674144
2	42.0	0.0	4.0	140.0	211.0	0.0	2.0	74.0	1.0	3.00082159845831
3	38.0	1.0	4.0	132.0	159.0	0.0	0.0	155.0	0.0	5.3990304222509 8
4	50.0	0.0	1.0	212.0	188.0	0.0	0.0	180.0	1.0	2.894363410327693
5	47.0	1.0	4.0	168.0	189.0	0.0	0.0	146.0	0.0	3.0539521019624285
6	48.0	1.0	1.0	185.0	195.0	0.0	0.0	125.0	0.0	1.5781421876502242
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	35.0	1.0	4.0	107.0	189.0	1.0	2.0	164.0	1.0	0.41993214564443815
2	33.0	0.0	4.0	80.0	367.0	0.0	2.0	172.0	0.0	1.8747085656655706
3	36.0	1.0	4.0	125.0	150.0	0.0	0.0	118.0	0.0	1.4691448683575297
4	35.0	1.0	4.0	154.0	193.0	0.0	0.0	162.0	0.0	0.6246588747371649
5	41.0	1.0	4.0	102.0	276.0	1.0	0.0	191.0	0.0	2.5847975492521784
6	49.0	0.0	3.0	79.0	90.0	1.0	2.0	201.0	0.0	1.1784187222302633
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	35.0	1.0	3.0	137.0	271.0	0.0	2.0	191.0	0.0	0.6791726773803732
2	43.0	1.0	4.0	161.0	329.0	0.0	2.0	154.0	1.0	0.5102504839369475
3	27.0	0.0	3.0	149.0	324.0	0.0	0.0	144.0	0.0	2.8470980466499736
4	42.0	1.0	4.0	158.0	212.0	0.0	0.0	179.0	1.0	0.4649775451235723
5	57.0	1.0	3.0	164.0	292.0	0.0	2.0	182.0	0.0	-0.028006805433103465
6	67.0	1.0	4.0	158.0	366.0	0.0	0.0	185.0	0.0	1.4507753529557592
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	70.0	1.0	2.0	115.0	353.0	0.0	0.0	149.0	0.0	-0.32908724890676
2	54.0	0.0	4.0	126.0	354.0	0.0	2.0	179.0	0.0	1.3535534603322645
3	69.0	1.0	3.0	114.0	295.0	1.0	2.0	194.0	0.0	0.009636774682052873
4	65.0	1.0	4.0	124.0	168.0	0.0	2.0	202.0	0.0	0.38917873193351915
5	53.0	0.0	4.0	118.0	379.0	0.0	2.0	138.0	0.0	-0.19287070503337153
6	69.0	1.0	1.0	106.0	369.0	0.0	0.0	193.0	0.0	-0.18913289263206753
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	64.0	1.0	2.0	150.0	202.0	0.0	2.0	209.0	0.0	1.7493594169266005
2	66.0	1.0	4.0	166.0	318.0	0.0	0.0	197.0	1.0	2.6659680615842936
3	74.0	1.0	2.0	206.0	341.0	0.0	0.0	161.0	1.0	2.8891332726763674
4	53.0	1.0	1.0	155.0	348.0	0.0	0.0	200.0	1.0	1.7617425108521716
5	67.0	1.0	1.0	161.0	345.0	0.0	2.0	148.0	0.0	-0.06210179426682
6	69.0	1.0	4.0	175.0	204.0	0.0	0.0	161.0	0.0	0.03502225323333086
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	51.0	0.0	3.0	145.0	419.0	0.0	0.0	122.0	1.0	0.23599803624632532
2	41.0	1.0	4.0	172.0	406.0	0.0	0.0	172.0	1.0	2.2060411291887245
3	43.0	1.0	2.0	175.0	372.0	0.0	0.0	186.0	1.0	0.6777573943802592
4	61.0	1.0	1.0	132.0	232.0	0.0	2.0	190.0	0.0	2.3085394422187866
5	37.0	0.0	4.0	148.0	317.0	0.0	0.0	165.0	0.0	1.985284577084522
6	46.0	1.0	2.0	153.0	367.0	1.0	0.0	137.0	0.0	2.3298012730583377
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	51.0	0.0	3.0	134.0	242.0	1.0	0.0	109.0	1.0	0.038466279767562166
2	49.0	1.0	4.0	187.0	282.0	0.0	2.0	199.0	0.0	2.044618553571543
3	78.0	1.0	4.0	148.0	264.0	0.0	2.0	157.0	0.0	-0.447513098971816
4	44.0	0.0	3.0	175.0	247.0	0.0	2.0	194.0	0.0	8.765984856772985
5	53.0	0.0	3.0	116.0	151.0	0.0	2.0	189.0	0.0	-1.1884500149529318
6	44.0	0.0	4.0	137.0	204.0	0.0	2.0	193.0	0.0	2.4215499668040916
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	49.0	1.0	1.0	167.0	243.0	0.0	0.0	146.0	1.0	1.0017509112121854
2	55.0	1.0	3.0	162.0	237.0	1.0	0.0	144.0	0.0	-0.2881271856068595
3	29.0	0.0	3.0	149.0	178.0	0.0	0.0	199.0	1.0	-0.4100797084558089
4	46.0	0.0	4.0	155.0	160.0	0.0	0.0	130.0	1.0	1.831608512424089
5	61.0	1.0	3.0	231.0	144.0	0.0	1.0	125.0	0.0	3.5801711154333478
6	48.0	1.0	3.0	167.0	123.0	0.0	2.0	145.0	1.0	1.0851034462114169
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	66.0	1.0	3.0	149.0	307.0	1.0	2.0	168.0	0.0	0.4515008723036945
2	61.0	1.0	1.0	144.0	242.0	0.0	0.0	129.0	1.0	2.2880978330892034
3	75.0	1.0	3.0	137.0	307.0	0.0	0.0	183.0	1.0	1.8493266986184373
4	53.0	1.0	1.0	151.0	263.0	0.0	2.0	176.0	0.0	1.8351535382678839
5	73.0	0.0	4.0	128.0	288.0	0.0	0.0	133.0	1.0	0.3273767701931038
6	62.0	1.0	4.0	144.0	226.0	0.0	2.0	138.0	1.0	1.8093820639624048
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6